Customer Segmentation: Exploratory Data Analysis

From Raw Records to Predictive Foundations

Dataset Introduction & Scale

53,503
30

- 53,503 distinct records spanning 30 diverse features.
- Key features include Age, Income Level, Policy Type, Risk Profile, and Segmentation Group.

The Business Motivation

- Identify distinct customer segments.
- Dynamically optimize targeted marketing campaigns.
- Accurately assess risk profiles across cohorts.

The Operational Challenge

- Real-world data complexity.
- Initial raw data contains noise, duplicates, and structural gaps.
- Requires rigorous cleaning and feature engineering to forge a reliable source of truth.

Data Processing Pipeline - Forging a Reliable Source of Truth

Raw Data
(~54,500 rows)

→

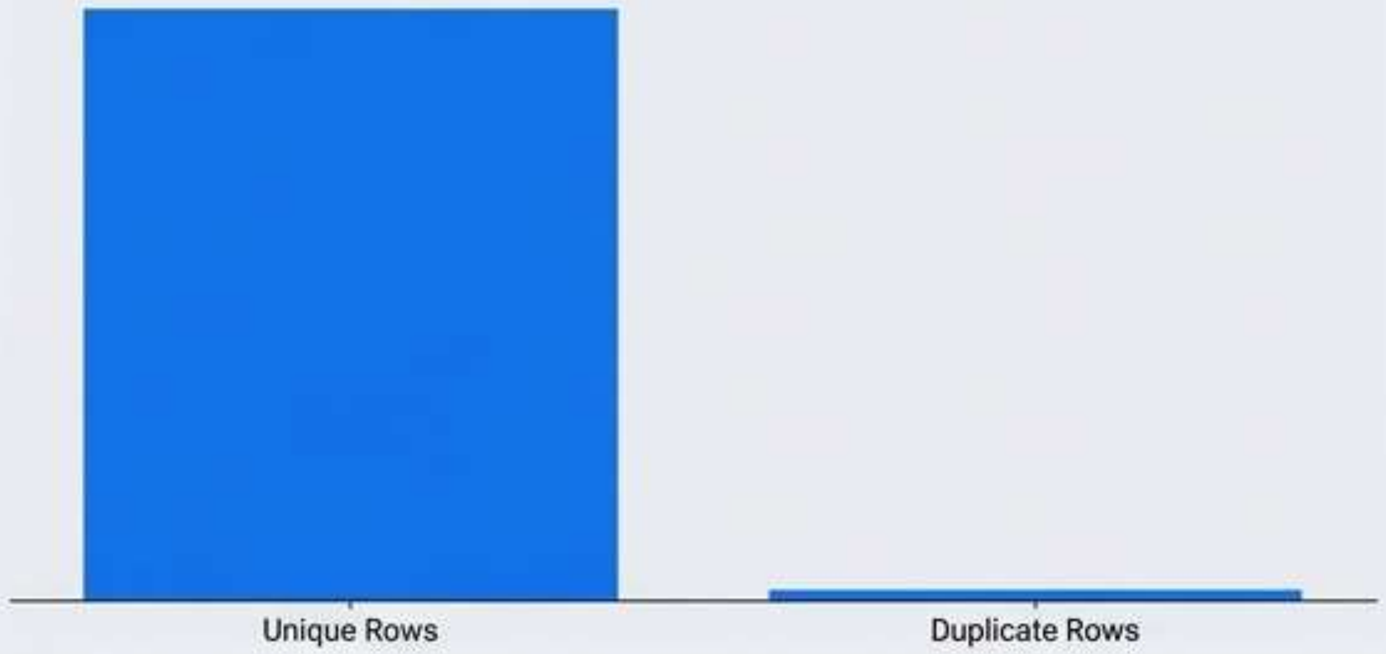
Filtering & Patching

→

Clean Data
(53,503 unique rows)

Removing Duplicates

- Purged redundant records, reducing the dataset from ~54,500 down to exactly 53,503 unique rows.
- Ensures strict analytical integrity.



Handling Missing Values

- Applied robust median imputation to preserve data distribution.
- Addressed critical gaps without allowing outliers to skew the baseline:
  - Age (~1000 missing)
  - Premium Amount (~500 missing)
  - Credit Score (~500 missing)
  - Income Level (~500 missing)

Feature Engineering - Unlocking the Dimension of Time

Policy Renewal Date (Text)

−

Policy Start Date (Text)

→

Calculated Policy Tenure (Days)

Date Conversion

- Transformed static, text-based date columns into dynamic datetime formats.
- Enables algorithmic interpretation of customer timelines.

Derived Behavioral Features

- Created Policy Tenure (Days) to establish a measurable metric for customer loyalty and lifecycle stage.
- Extracted Policy Start Year and Month to enable macro-trend analysis, cohort tracking, and seasonal pattern recognition.

Exploratory Data Analysis: The Customer Mix

Key Insights

- Segment 5 is our dominant cohort, representing the largest fraction of the customer base with ~14,000 customers.
- Segment 1 represents our smallest footprint, capturing only ~8,600 customers.
- Strategic Takeaway: Market penetration is currently heavily skewed toward Segment 5. This indicates either a strong product-market fit for this specific demographic or a systemic gap in capturing Segment 1 customers.

NotebookLM

Customer Segmentation Interpretation Matrix

Segment 1 | Budget Customers
Characteristics: Lower income, lower premium, moderate credit score.
Business Meaning: Basic policies.

Segment 2 | Standard Customers
Characteristics: Moderate income and premium, stable credit score.
Business Meaning: Balanced risk and value.

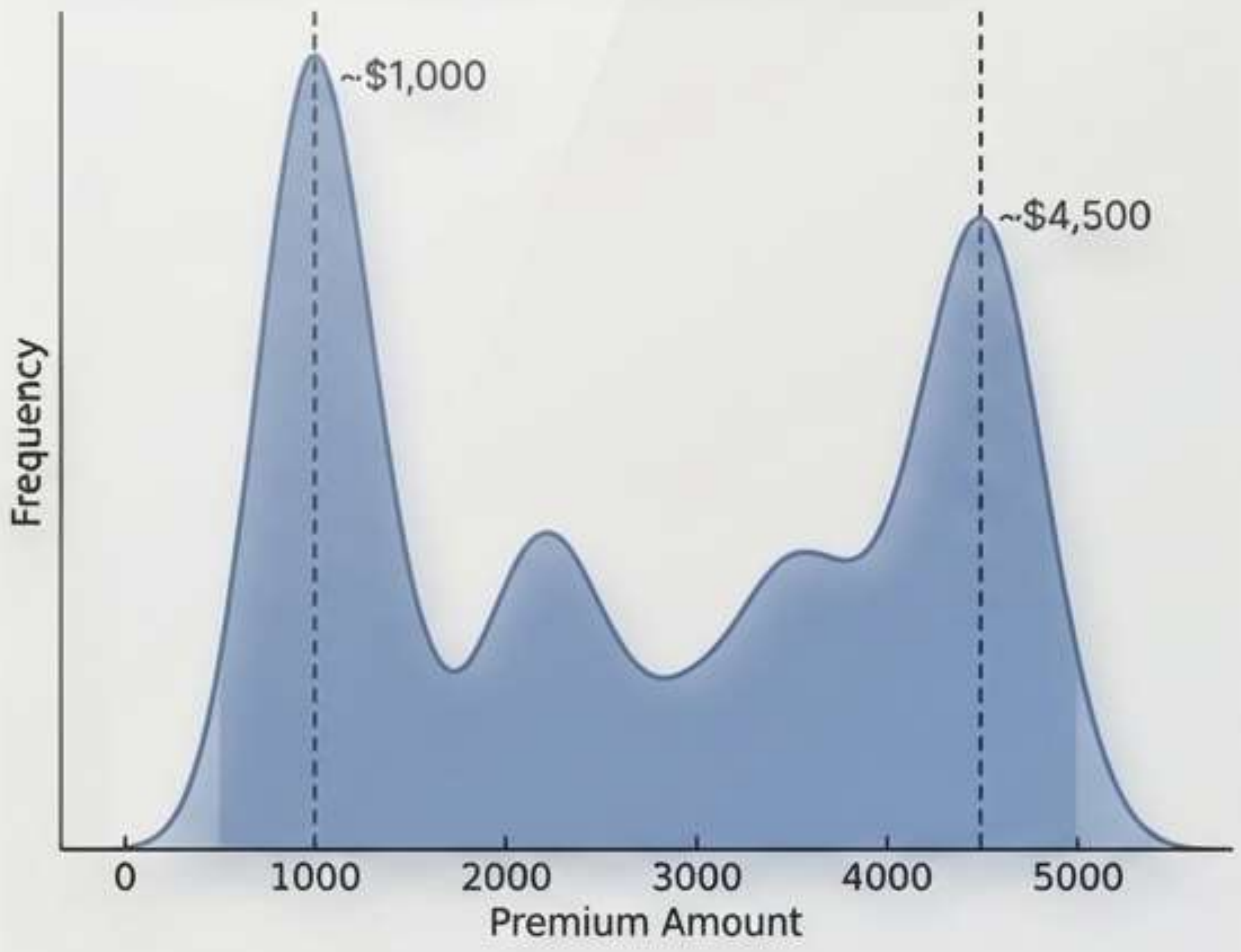
Segment 3 | High-Value Customers
Characteristics: Higher income, higher premium, strong credit score.
Business Meaning: Core revenue contributors.

Segment 4 | Riskier Customers
Characteristics: Higher risk profile, lower credit score, possible claims history.
Business Meaning: Requires careful underwriting.

Segment 5 | General Customer Base
Characteristics: Mixed characteristics.
Business Meaning: Largest group with highly diverse profiles.

Exploratory Data Analysis: The Financial Footprint

Premium Amount Distribution

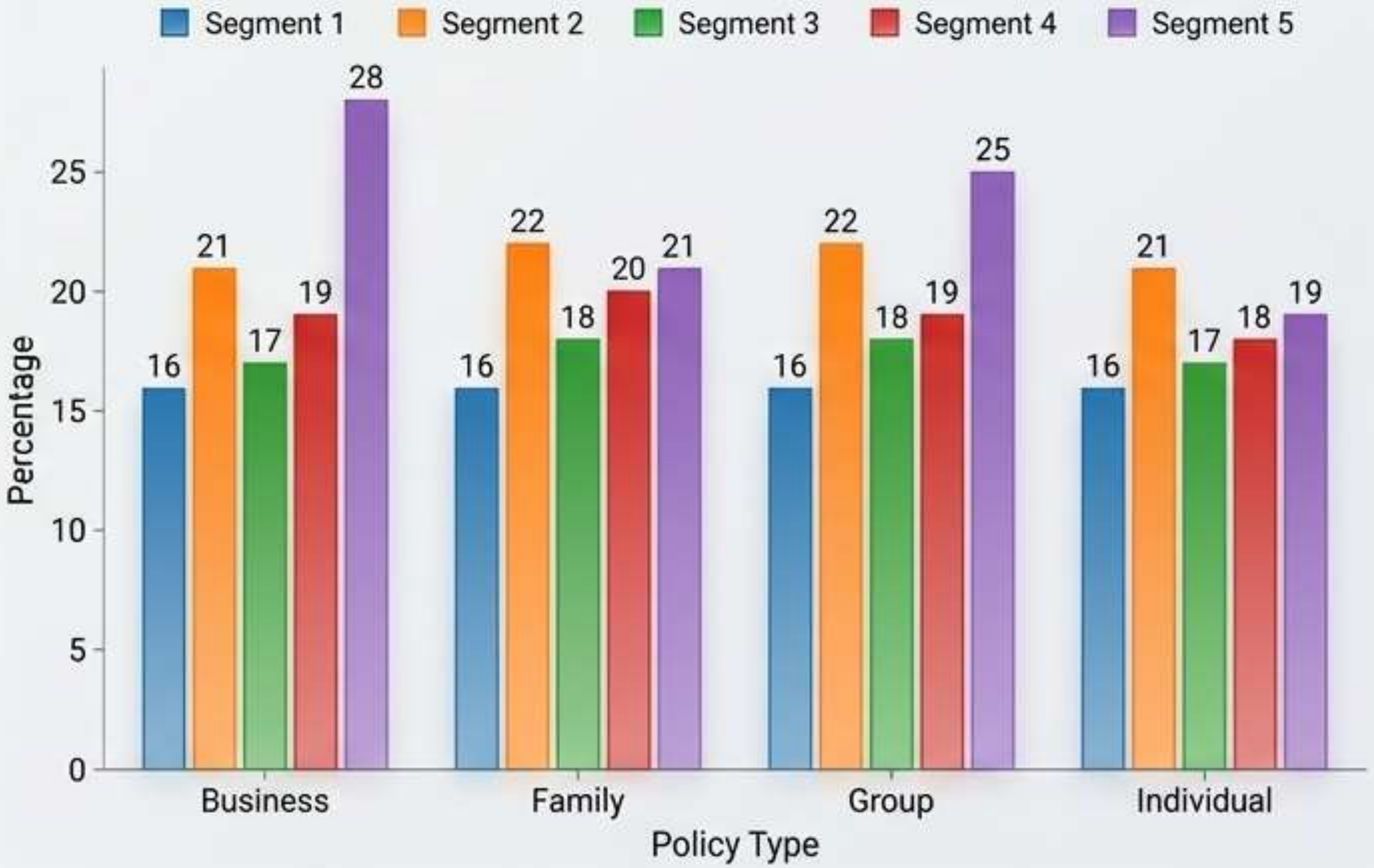


Portfolio Breakdown

- **Multi-Modal Pricing Architecture:** The distribution of Premium Amounts does not form a smooth bell curve. Instead, it features several distinct, recurring peaks.
- **Volume Clustering:** We observe massive customer clustering at specific price floors (e.g., ~$1,000) and premium ceilings (e.g., ~$4,500).
- **Strategic Takeaway:** These distinct peaks strongly suggest our revenue is driven by standardized tier pricing, base policy minimums, and distinct product bundles rather than highly customized, sliding-scale pricing.

Synthesis Matrix: Segment Preferences by Policy Type

Percentage Distribution of Segments within Policy Types



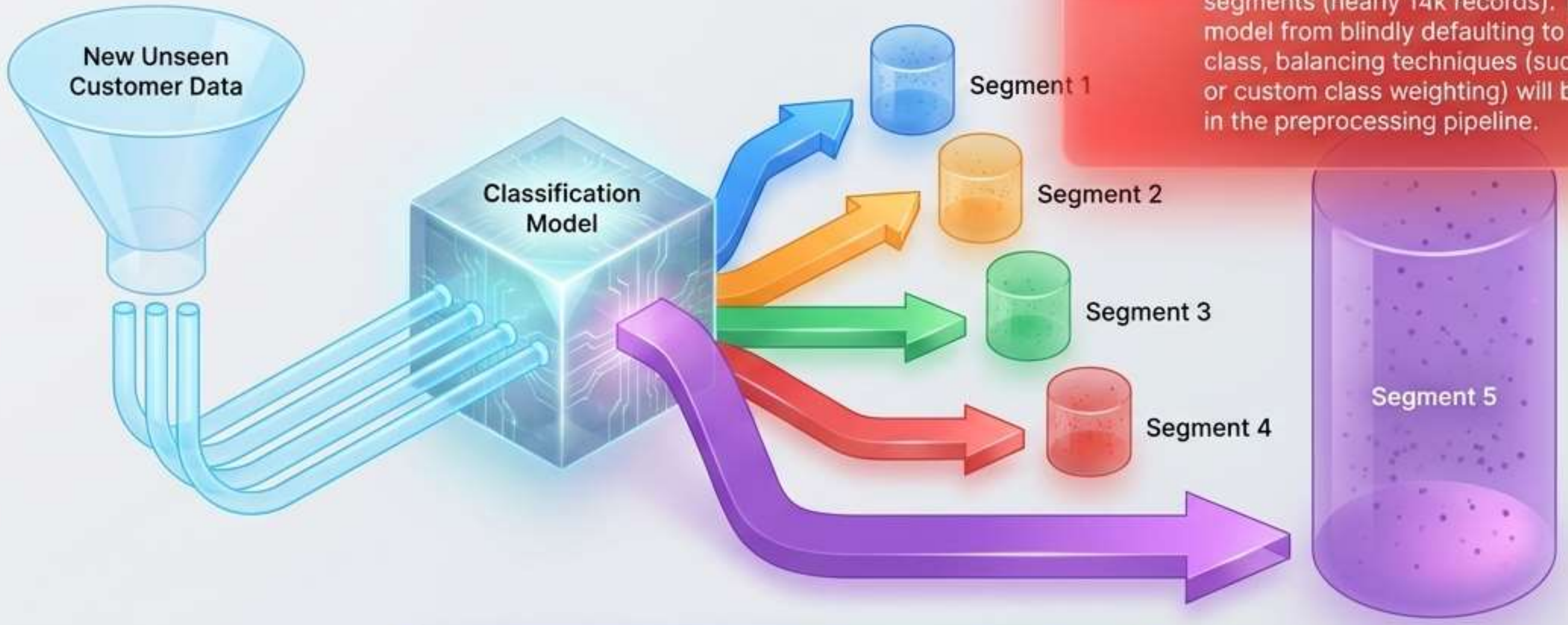
Strategic Insights

- **Non-Uniform Purchasing:** Customer segments exhibit distinct, measurable biases toward specific product lines.
- **Segment 5 Profile:** Shows a highly pronounced native preference for Individual and Group policies.
- **Segment 2 Profile:** Markedly over-represented in Business and Family policies compared to all other cohorts.
- **Strategic Takeaway:** Campaign spend can be aggressively optimized by routing specific Policy Type marketing collateral directly to the segments that natively prefer them.

Predictive Modeling Pipeline: Supervised Classification

Class Imbalance Warning

Segment 5 drastically outweighs the other segments (nearly 14k records). To prevent the model from blindly defaulting to the majority class, balancing techniques (such as SMOTE or custom class weighting) will be mandatory in the preprocessing pipeline.



Proposed Next Step: Train a supervised classification model to automatically predict segment placement for new policyholders based on their foundational traits.

Summary Roadmap & Next Steps

Phase 1 Completed: Data Prep

Executed rigorous data cleaning, median imputation, and datetime feature engineering across 53,000+ records.

Phase 2 Completed: EDA & Insights

Uncovered distinct multi-modal
Uncovered distinct multi-modal
pricing, isolated segment-policy preferences, and mapped quantifiable risk-premium scaling.

Phase 3 In Progress: Model Training

Initiating development of the Multi-class Classification pipeline, integrating class-weighting protocols to neutralize target imbalance.

Open for Questions & Discussion.

Group 8: The Data Refinery Team

Prathmesh Mankar

Focus Area: Data Preparation, Pipeline Architecture, & Dashboarding.

(Driven by median imputation, duplicate resolution, and initial dashboard setup)

Kshitija Shinde

Focus Area: Feature Engineering & Time Dimension.

(Calculated policy tenure and behavioral feature extraction)

Parth Patel

Focus Area: Exploratory Data Analysis & Synthesis.

(Uncovered the interpretation matrix and multi-modal financial footprint)

Vrishin Dharmesh Kunnatham Parambath

Focus Area: Predictive Modeling & Dashboarding.

(Supervised classification setup and class-weighting protocols)

Based on Slide C

Project Resources

Interactive Segment Dashboard

Explore the multi-modal pricing and segment preferences in real-time.

https://public.tableau.com/app/profile/kshitija.dipakrao.shinde/viz/CustomerRiskRevenueIntelligenceDashboard/CustomerRiskRevenueIntelligenceDashboard?publish=yes

Source Code & Modeling Pipeline

Review the data cleaning scripts, EDA notebooks, and SMOTE implementation.

https://github.com/Kshitija-Shinde9/Insurance-Claims-Analysis

Transparency in process. Confidence in prediction.